Arbitrary Spike Time Dependent Plasticity (STDP) in Memristor by Analog Waveform Engineering

Neeraj Panwar, Bipin Rajendran, and Udayan Ganguly

***Abstract*— In the literature, various *pulse*-based programming schemes have been used to mimic *typical* spike time-dependent plasticity (STDP)-based learning rule observed in biological synapses. In this letter, we demonstrate the capability to generated *arbitrary* STDP behaviors by using analog programming waveforms inspired by neuronal action potential. First, we propose a simple algorithm to generate any arbitrary form of STDP. Second, we show the feasibility of a range of spike correlation time scales for STDP, e.g., biological (∼100 ms) to accelerated (∼20μs), based on W/$Pr_{0.7}Ca_{0.3}MnO_3$/Pt based memristor. Third, we experimentally demonstrate several forms of STDP behaviors, where the pre- and post-neuronal waveforms are randomly spaced in time, akin to operational conditions. STDP shape corresponds well to waveforms. Thus, we show that artificial synapses can achieve the richness observed in biology as well as a range of STDP timescales for biologically compatible to accelerated neural network applications.**

***Index Terms*— PCMO, exponential waveform, RRAM.**

I. Introduction

TO BUILD large computational systems implementing brain-inspired machine learning algorithms for data analytics applications, nanoscale devices mimicking the dynamics of neurons and synapses of the brain are necessary. The basic spiking neural network (SNN) architecture consists of a set of pre-synaptic (input) neurons connected to a set of post-synaptic (output) neurons through a cross-point array of synapses represented by memristors of conductance G (Fig. 1(a)). Learning is equivalent to re-routing the input spikes to different output neurons by modifying the conductivity (e.g. open vs. short) of the synapses. In biology, the learning rule to modify the synaptic conductivity is called spike timing dependent plasticity (STDP), which is described below. Fig. 1 (b) shows a pre-neuron connected to a post-neuron through a synapse represented by a conductance (G). The pre-neuron and post-neuron spike at time t_{pre} and t_{post} respectively, while time- correlation is $\Delta t = t_{post} - t_{pre}$. In biological synapses, the conductance change (ΔG) of the synapse depends upon Δt (Fig. 1 (c)). When pre-neuron spikes before the post-neuron (i.e. $\Delta t > 0$), then the spike pair is considered causal i.e. pre-neuron caused post-neuron to spike. On the other hand, when pre-neuron spikes after the post-neuron (i.e. $\Delta t < 0$), then the spiking is considered anti-causal i.e. pre-neuron did not cause the post-neuron to spike. The specific STDP (Fig. 1(c)) produces a learning rule where causal spikes increase synaptic conductance (i.e. long term potentiation or LTP) while anti-causal spiking decrease synaptic conductance (i.e. long term depression or LTD) to modify the network connectivity. As the time-correlation decreases (i.e. $|\Delta t|$ increases), conductance change (i.e. $|\Delta G|$) is weaker. The implementation of STDP in nanoscale devices is a central challenge in neuromorphic engineering. It requires analog change in resistance of the synapse.

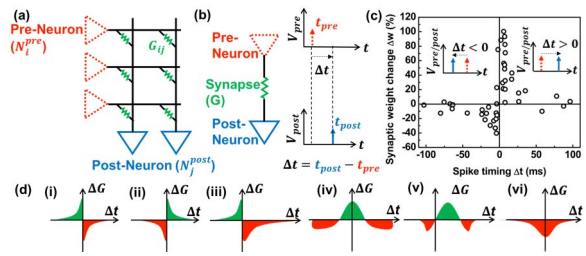

Fig. 1. (a) A SNN architecture with pre-neurons (N_i^{pre}) connected to post-neurons (N_j^{post}) through a synapse (with conductance G_{ij}). (b) A single pre- and post-neuron spike at t_{pre} and t_{post} respectively such that $\Delta t = t_{post} - t_{pre}$. (c) Biological synaptic weight change in hippocampal glutamatergic synapse measured by Bi and Poo. Figure taken from [1]. (d) Various synaptic plasticity observed in biology [2], (i) neocortex-layer 5 Xenopus tectum hippocampus, (ii) ELL of electric fish, (iii) neocortex-layer 2/3 hippocampus, (iv) GABA-ergic neurons in hippocampal culture, (v) STDP at CA3-CA1 synapse [3] (vi) neocortex-layer 4 spiny stellates indicates that the nature of STDP is a critical engineering handle in SNN.

There have been 5 different approaches of STDP implementation using resistive random access memories (RRAM). First, to circumvent the *abrupt* reset (i.e. LTD) process in

Manuscript received April 6, 2017; accepted April 17, 2017. Date of publication April 19, 2017; date of current version May 22, 2017. This work was supported in part by the Science and Engineering Research Board, in part by the Department of Science and Technology, and in part by the Department of Electronics and IT, Government of India. The review of this letter was arranged by Editor M. Radosavljevic. *(Corresponding author: Neeraj Panwar.)*

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Digital Object Identifier 10.1109/LED.2017.2696023

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phase change memory (PCM) devices, a 2-PCM synapse has been demonstrated where the gradual set process of two different PCM devices is used to produce *gradual* LTP and LTD by current comparison through a differential amplifier to demonstrate a 2-level *digital* STDP [4]. Second, a complex pre-post neuron digital signaling protocol has been proposed to implement a *multi-level* STDP [5], [6], which requires a *global bus* for communication and circuits to compute Δt for every synapse and consequent ΔG modulation. This approach has a high power, circuit area penalty [7]. Third, a *local implementation of analog STDP* involves the superposition of two waveforms from pre- and post-neurons to produce a *pulse-width (t_{pulse}) modulation* [8], [9]. It is shown that t_{pulse} depends upon Δt exponentially (i.e. $t_{pulse} = \lambda \exp(-\Delta t/\tau)$) e.g. $t_{pulse} = 200ms$ is required to implement Δt of 10ms while $t_{pulse} = 50ms$ is required to implement Δt of 30ms [8]. The observation that t_{pulse} exceeds Δt constrains *real-time learning* using SNN. Fourth, an alternative local implementation of *multi-level* STDP involving the superposition of two pulse-train based waveforms from pre- and post- neurons has been demonstrated to produce different Δt dependent effective *pulse-height* (V_{pulse})*modulation*. [10]–[13]. A variant using a transistor as a selection device at every synapse has been explored [5], [14]. However, it has been shown that analog waveforms are preferred from energy and circuit area perspective [7]. We have recently demonstrated two standard forms of STDP using *analog waveforms* in $Pr_{0.7}Ca_{0.3}MnO_3$(PCMO) based RRAM [15]. In biology, various different STDP rules are observed as shown in Fig. 1 (d) [2], [3], which indicates that the control of the nature of STDP is an essential enabler for SNN, which is not yet demonstrated in artificial synapses.

In this letter, we use PCMO based RRAM to demonstrate a general method of implementing an arbitrary STDP. First, an approximately linear G vs. V_{pulse} dependence is demonstrated. Second, a simple method of generating pre- and post-neuron waveform to produce an arbitrary STDP is shown. Thirdly, waveform based typical STDP is demonstrated over a range of correlation timescales (Δt) to mimic biology and enable hardware acceleration. Finally, various arbitrary STDP rules are demonstrated experimentally.

II. Experimental

The devices were fabricated on 4″ silicon wafer. After wafer was cleaned with 2% HF dip for 30s, 30nm SiO_2 was grown by rapid thermal oxidation. This was followed by bottom electrode (BE) formation, using sputter deposition of Ti/Pt (9nm Ti adhesion layer followed by 57nm Pt). PCMO film was then deposited by pulsed laser deposition in 1.3×10^{-2}mbar O_2 partial pressure at room temperature. The sample was annealed in rapid thermal anneal for 120s in inert (N_2) ambient at 650° C. A 5μm diameter gold shielded Tungsten tip was used as the top contact as shown in literature [16] which is demonstrated to be fast compared to microfabricated pads [17] and highly repeatable [18]. DC I-V sweep measurements were carried out by Agilent B1517A Source Measurement Unit (SMU) while all the pulse and waveform based measurements were carried out by Agilent B1530A Waveform Generator/Fast Measurement Unit (WGFMU).

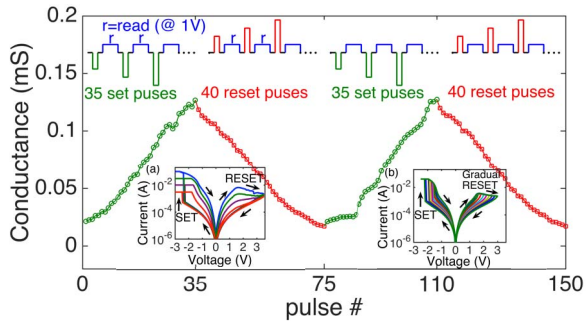

Fig. 2. The analog conductance by pulse programming in PCMO device is shown. (Insets show the DC I-V sweep characteristics of the PCMO device. Inset (a). The set resistance can be controlled by the current compliance level. Inset (b). while reset resistance can be controlled by the reset voltage).

III. Results and Discussion

PCMO based RRAM has been extensively investigated due to forming-less, low variability, excellent retention and endurance properties [19]–[22]. Current conduction by trap space charge limited current has been widely reported which depends upon trap density change [23]–[25]. Traps are oxygen vacancy (V_O^{2+}) and trap density is modulated by V_O^{2+} drift [20], [26], [27]. Reactive metal-PCMO interface has an ultra-thin (hence relatively electrically conducting) sub-stoichiometric metal oxide (e.g. WO_x) layer that acts as a source/sink of V_O^{2+} [28]–[30]. Positive bias on W-electrode causes V_O^{2+} drift from the reactive (W) electrode into PCMO to increase trap density and reduce current i.e. conductance. Negative bias attracts V_O^{2+} out of PCMO to reduce trap density and increase current & conductance [28]–[30]. We have recently demonstrated that Joule heating during set/reset assists V_O^{2+} drift [17].

Set and reset in analog resistance states, by short pulse application are shown in Fig. 2. DC I-V sweep characteristics of the device are shown in insets. During sweep, set resistance can be controlled by the current compliance level during set (inset (a)) while reset resistance can be controlled by the reset voltage (inset (b)). Note that during sweep for low voltages (e.g. below ±1.2V), there is no change in the conductivity of the device, which is a threshold for set/reset operation. Linearly varying analog resistance states are achieved by 35 pulses of 4μs of linearly varying voltages from −1.600V to -2.125V with an increment of 15.4mV for set and 40 pulses of 10μs from +1.100V to +2.100V with an increment of 25.6mV for reset, (Fig. 2). A quasi-linear dependence of G vs. V_{pulse} is observed.

To generate any arbitrary STDP (e.g. shown in Fig. 3(a)), we now describe a general approach for designing the required pre- and post-neuron waveforms. First, since G is proportional to V_{pulse} (Fig. 2), the STDP waveform (ΔG vs. Δt) is converted to a Voltage waveform (i.e. V vs. t) (Fig. 3(b)). Then, the V-waveform is split into V>0 (green) denoting LTP and V<0 (red) denoting LTD. A "sharp" negative and positive spike ($t_{spike} = 5ms$) of magnitude V_{spike} of V_{T-} and V_{T+} is added to create the pre-neuron (red) and post-neuron (green) waveforms respectively, where $V_T+ = 1.2\ V$ and $V_{T-} = -1.2\ V$. The post-neuron (green) waveform

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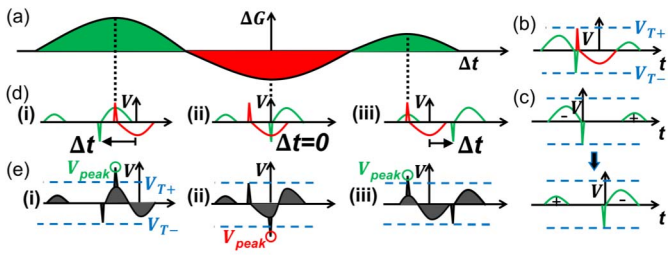

Fig. 3. Scheme to mimic (a) arbitrary spike timing dependent plasticity in PCMO device with potentiation (green) and depression (red) of conductance. (b) Decomposition of STDP waveform into positive and negative part as pre (red) and post (green) neuronal waveforms. (c) Post-neuron waveform is produced by taking (i) the positive STDP waveform and (ii) flipping in time. (d) Pre- and Post-neuronal waveforms offset to various extents in time (Δt) i.e. (i) negative ($\Delta t < 0$), (ii) $\Delta t = 0$, (iii) positive ($\Delta t > 0$) and (e) resultant superposition produces peak voltage $V_{peak}(\Delta t)$ which resembles STDP waveform. The $V_{peak}(\Delta t)$ along with the quasi-linear $G(V_{\text{peak}})$ dependence in Fig. 2, produces STDP (i.e. $\Delta G(\Delta t)$) as shown in (a).

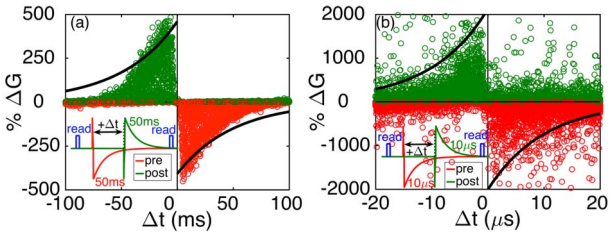

Fig. 4. Applied pre- and post-neuronal waveform (inset) produces standard anti-STDP for a range of time correlation (Δt), e.g. (a) 100ms ($\sim$ biological timescales) (b) 20μs (accelerated timescale) is demonstrated. Waveform equivalent line (WEL- black line) describes envelope of experimental STDP.

is flipped in time (Fig. 3 (c)). Δt offsets the pre- and post-neuronal waveforms (Fig. 3(d)). The resultant waveform ($V(t) = V_{post}(t) - V_{pre}(t)$) appears across the PCMO RRAM (Fig. 3(e)). The peak voltage (V_{peak}) vs. Δt follows the shape of the STDP waveform. When $|V_{peak}|$ exceeds resistance switching threshold ($V_{T\pm}$), then set or reset occurs. Due to the approx. linear G vs. V_{pulse} (Fig. 2), the STDP is produced.

Fig. 4 shows the experimental demonstration of standard anti-STDP by applying pre- and post- neuron waveforms. The Δt between pre- and post-neuron waveform pair are *randomly* chosen – akin to operational conditions. Since the RRAM device is not perturbed by voltages below ± 1.2V, a 0.5V read pulse before and after application of waveforms is recorded to measure ΔG. A correlation time (Δt) of ± 100 ms, akin to biology, is implemented (Fig. 4 (a)). Percentage conductance change ($\%\Delta G$) is calculated [5] as

$$\%\Delta G = \frac{(G_f - G_i)}{\min(G_i, G_f)} \times 100$$

where, G_i is the initial conductance before applying nth STDP cycle, G_f final conductance after applying nth STDP cycle and $\min(G_i, G_f)$ is the minimum of G_i, G_f.

A set of >3000 random pre- and post-neuron waveform-pairs are plotted. To compare the STDP to the waveforms, waveform-equivalent-line (WEL) e.g. $\Delta G = Aexp(-\Delta t/\tau)$

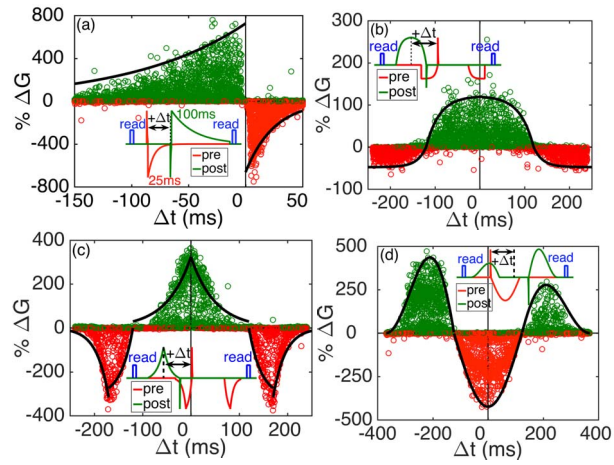

Fig. 5. Applied pre- and post-neuronal waveform (inset) produces various timing dependent plasticity behaviors (a) anti-STDP with modulated time window, (b) STDP observed in synapses of GABA-ergic neuron in hippocampal culture, (c) symmetric STDP with potentiation at $\Delta t \sim 0$ like [31] and (d) an arbitrary asymmetric STDP. Waveform equivalent line (WEL- black line) describes envelope of the experimental STDP.

is plotted for an exponential function, where τ is the decay time constant. The pre-factor $A = \Delta G(\Delta t = 0)$ is determined experimentally by generating a cumulative distribution function (cdf) of all ΔG data points and using the 99.9% value as A. We see that WEL envelopes STDP data. Hence, the waveform timescale describes the STDP well (Fig. 4(a)). We observe that the switching time-scale of device limits the lower limit of the spike time (i.e. $t_{pulse} \ll \Delta t$) of the waveform. PCMO RRAM has a faster switching timescale of $t_{pulse} \sim 100$ns [17]. We were also able to show the STDP where t_{spike} is $1\mu s$ while the waveform time-difference i.e. Δt range was $\pm 20\mu s$ (Fig. 4(b)). Thus, a range of Δt from 100ms (biological) to $20\mu s$ (accelerated hardware timescale) can be engineered. In Fig. 5(a-d), we show that arbitrary STDP can be generated with *deterministic* shape with excellent correspondence to the waveform.

IV. Conclusions

Analog STDP is considerd to be efficient in energy and area. In this letter, we have demonstrated that PCMO device is able to show a quasi-linear relationship between analog conductance change and applied programming pulse voltage magnitude. Leveraging this characteristic feature of PCMO, we have proposed a scheme that generates arbitrary STDP behaviors. Both biological and accelerated timescale is experimentally demonstrated staying true to real-life operational conditions. Various STDP are demonstrated where the shape of the STDP is strongly correlated to the designed neuronal V-waveform. Thus, the richness of biological synapse is enabled in artificial synapses by the arbitrary STDP implementation process.

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